MANIN'S CRITERION FOR AN ISOMORPHISM OF MOTIVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to arbitrary objects through the Karoubian envelope and the twist, the translation into morphism groups out of the unit, and the final use of left duals and the Yoneda lemma.

Editorial note. The label lemma-manin names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-manin –start 1326 –end 1356 –local-id 28149 returned RESULT=OK (L28149 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Objects of the category of motives are summands of twisted motives of schemes. The construction of the category of motives as the Karoubian envelope of the category of correspondences, so that every object is a summand of a twisted motive of a smooth projective scheme. Reduces the assumption from schemes to arbitrary objects, using that taking Chow groups commutes with the twist up to a shift.
- Chow groups as morphism groups from the unit. Same chapter, the lemma representing Chow groups as morphism groups out of the unit. Converts the isomorphism on Chow groups into an isomorphism of morphism groups out of the unit after tensoring with any object.
- Existence of left duals and the Yoneda lemma. Same chapter, the lemma that every object of the category of motives has a left dual, and the categorical lemma that a morphism inducing isomorphisms on all morphism groups is an isomorphism. Upgrades the isomorphism with the unit on the left to an isomorphism of all morphism groups and concludes by the Yoneda lemma.

Source reference. Original lemma 0FFQ: sources/stacks-chow/tags.txt, source lines 19295–19295. Prerequisite pointer invoked at proof line(s) 1353; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:weil.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 001P: sources/stacks-chow/tags.txt, source lines 70–70. Prerequisite pointer invoked at proof line(s) 1355; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FGM: sources/stacks-weil/pinned-original-source.tex, source lines 1306–1306. Prerequisite pointer invoked at proof line(s) 1341; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FGJ: sources/stacks-weil/pinned-original-source.tex, source lines 1238–1238. Prerequisite pointer invoked at proof line(s) 1348; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Manin)**.** *Let k be a base field. Let $c : M \to N$ be a morphism of motives. If for every smooth projective scheme X over k the map $c \otimes 1 : M \otimes h(X) \to N \otimes h(X)$ induces an isomorphism on Chow groups, then c is an isomorphism.*

Proof. Any object L of M_k is a summand of $h(X)(m)$ for some smooth projective scheme X over k and some $m \in \mathbf{Z}$. Observe that the Chow groups of $M \otimes h(X)(m)$ are the same as the Chow groups of $M \otimes h(X)$ up to a shift in degrees. Hence our assumption implies that $c \otimes 1 : M \otimes L \to N \otimes L$ induces an isomorphism on Chow groups for every object L of M_k. By Lemma 0FGM we see that

$$\mathrm{Hom}_{M_k}(\mathbf{1}, M \otimes L) \to \mathrm{Hom}_{M_k}(\mathbf{1}, N \otimes L)$$

is an isomorphism for every L. Since every object of M_k has a left dual (Lemma 0FGJ) we conclude that

$$\mathrm{Hom}_{M_k}(K, M) \to \mathrm{Hom}_{M_k}(K, N)$$

is an isomorphism for every object K of M_k, see Categories, Lemma 0FFQ. We conclude by the Yoneda lemma (Categories, Lemma 001P). □